Data Engineering Lab

SQL, Linux, AWS and Azure Practical Record

Name: Yedoti Premchand

Program: B.Tech Computer Science Engineering

University: Jain (Deemed-to-be University), Bengaluru

SQL Query to Find the Second Highest Salary

To find the second highest salary from the employees table using MySQL.

- Created the employees table (emp_id, emp_name, dept_name, salary) and inserted 8 records.
- Used a subquery: the maximum salary that is less than the overall maximum salary.
- Highest salary is 80000, so the second highest salary is 75000.

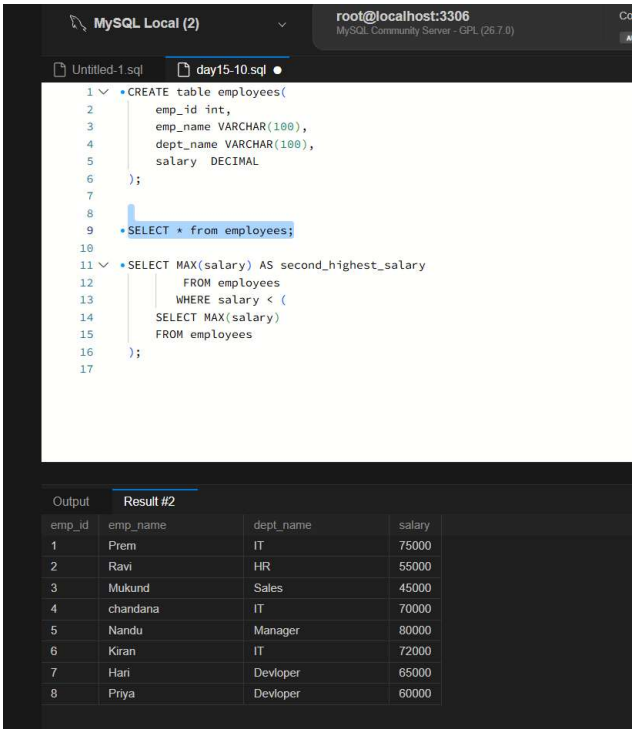


Figure 1.1: Table creation, data and query in MySQL Workbench

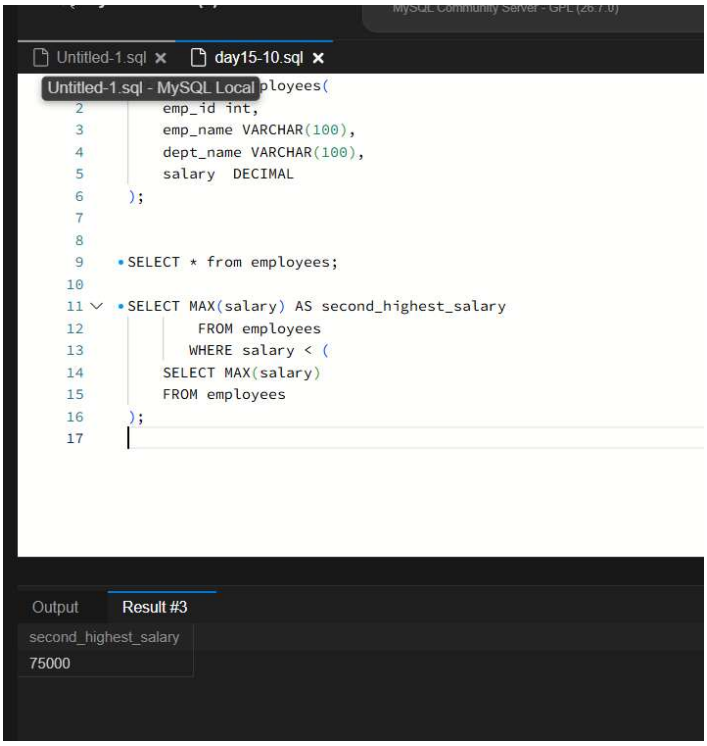


Figure 1.2: Query output, second_highest_salary = 75000

Linux Commands

To show the current working directory, create a directory, create two files, and list the files with their path.

- pwd shows the current directory; mkdir creates the directory; touch creates two files.
- ls -l lists the files with permissions; cd moves into the directory and pwd confirms the path.

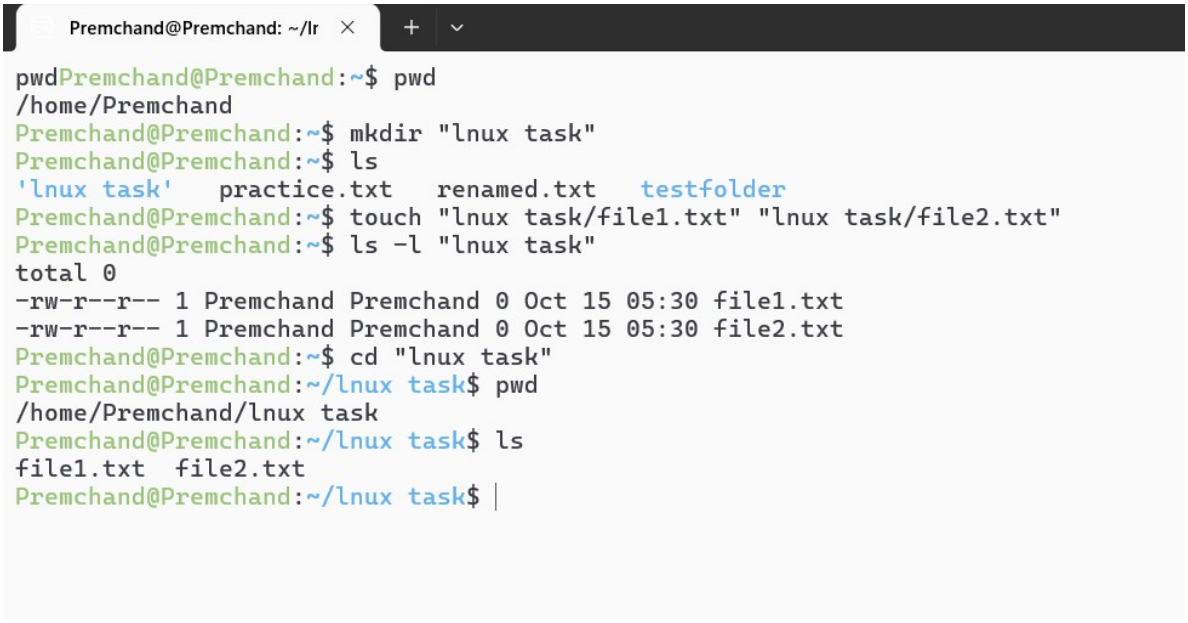


Figure 2.1: Linux terminal output

AWS EC2 Instance Details

To check the current state, instance type and public IP address of an EC2 instance.

- Instance name: dat15-10
- Instance state: Running
- Instance type: t2.micro
- Availability zone: us-east-1d
- Public IPv4 address: 34.207.250.147

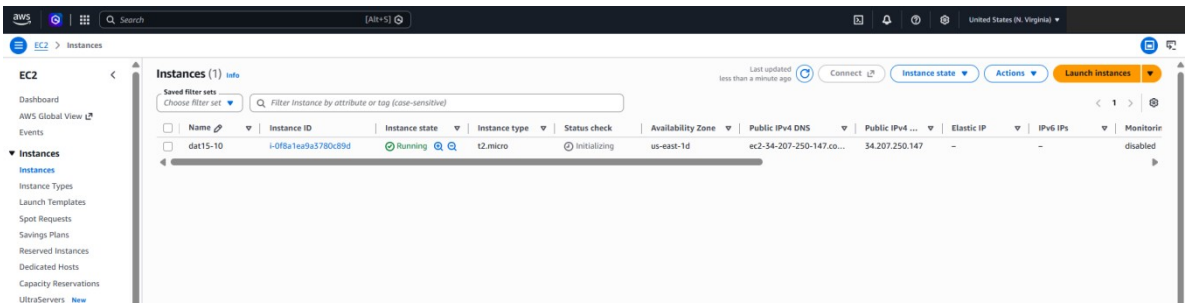


Figure 3.1: EC2 instance details

AWS IAM Role with S3 Read-Only Permission

To create an IAM role for S3 and add the S3 read-only permission.

- Created the role day15-10 with AWS service S3 as the trusted entity.
- Attached the S3 read-only permission policy to the role.

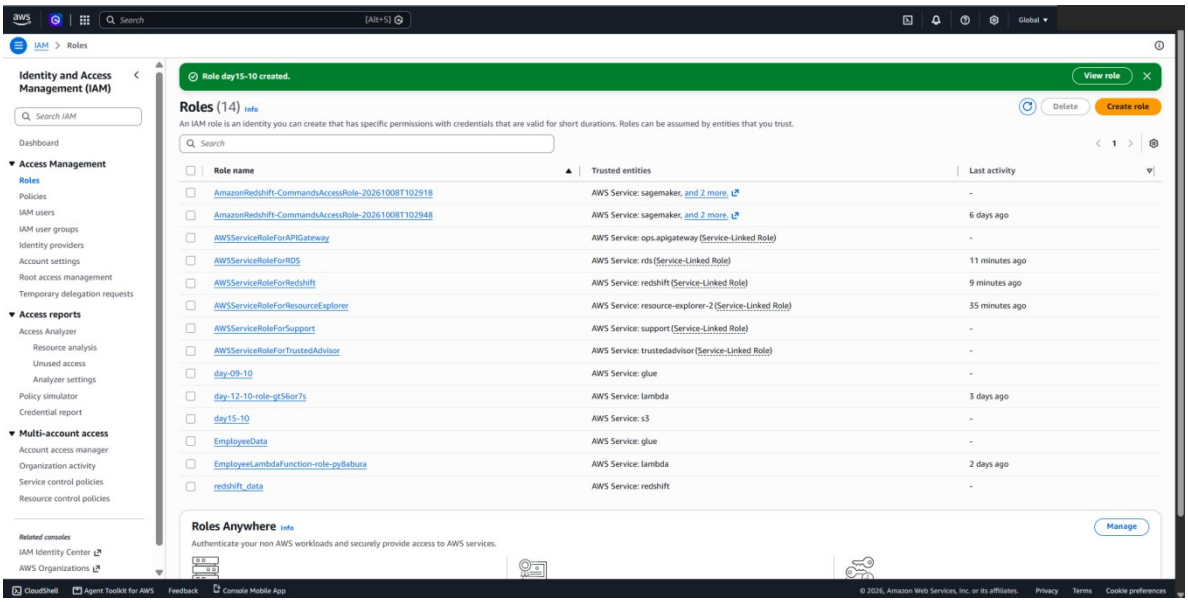


Figure 4.1: IAM role created successfully

Upload a File to an S3 Bucket

To upload a file to an Amazon S3 bucket.

- Bucket: day15-10
- File uploaded: customer_records.csv (text/csv, 196 B)
- Status: Upload succeeded

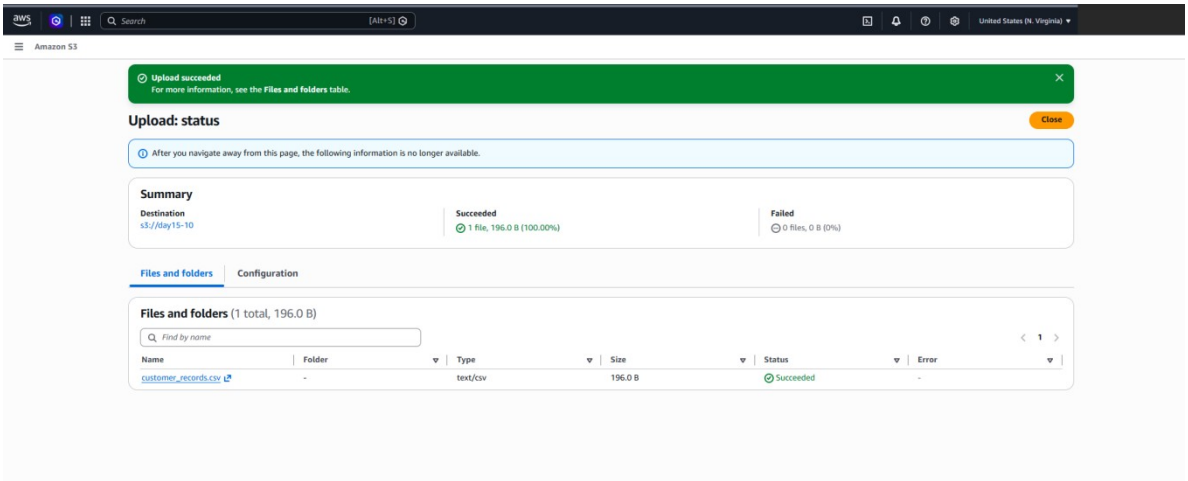


Figure 5.1: S3 upload status

Azure Resource Group

To create a resource group in Microsoft Azure.

- Resource group name: ParcticalRG
- Subscription: Azure for Students
- Region: East Asia

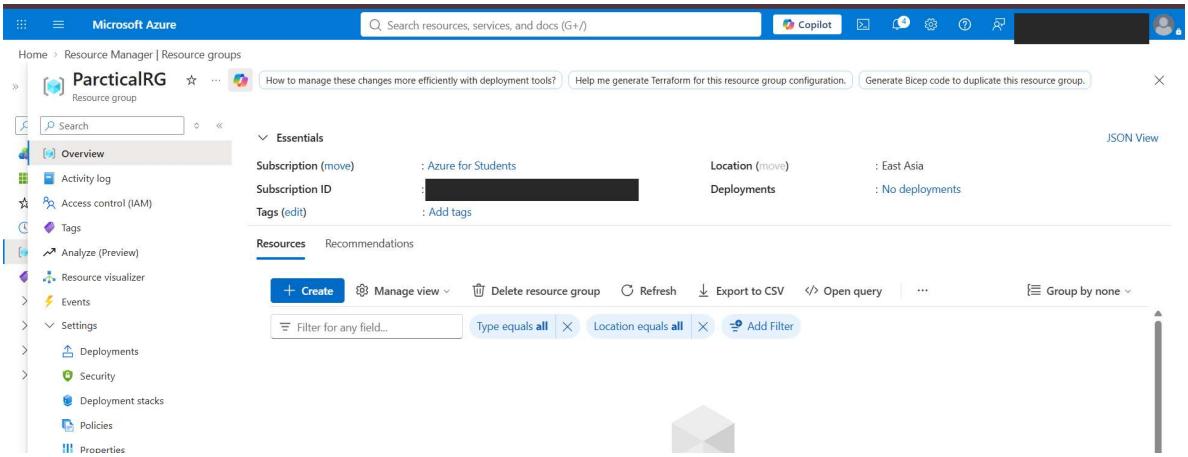


Figure 6.1: Azure resource group overview